🔒 Security in Computing

Packet Tracer Practicals – Complete Command Reference

University of Mumbai · BSc IT · Teacher's Reference Manual (2018–19)

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Practical 1 – Configure Cisco Routers for Syslog, NTP, and SSH Operations

Routers: R1, R2, R3 · PCs: PC-A, PC-B, PC-C · Servers: NTP Server (192.168.1.5), Syslog Server (192.168.1.6)

Objectives

- Configure OSPF MD5 Authentication
- Configure NTP (with Authentication)
- Configure Syslog Logging
- Configure SSH on R3

Part 1 – Configure OSPF MD5 Authentication

Step 2 – Enable OSPF MD5 Authentication (Area 0)

R1

```
R1(config)# router ospf 1
R1(config-router)# area 0 authentication message-digest
```

R2

```
R2(config)# router ospf 1
R2(config-router)# area 0 authentication message-digest
```

R3

```
R3(config)# router ospf 1
R3(config-router)# area 0 authentication message-digest
```

Step 3 – Configure MD5 Key on Serial Interfaces (Key 1 · Password: MD5pa55)

R1

```
R1(config)# interface s0/0/0
R1(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
```

R2

```
R2(config)# interface s0/0/0
R2(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
R2(config-if)# interface s0/0/1
R2(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
```

R3

```
R3(config)# interface s0/0/1
R3(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
```

Step 4 – Verify

R1 / R2 / R3

```
show ip ospf interface
```

Part 2 – Configure NTP

Step 1 – Enable NTP Authentication on PC-A (NTP Server)

PC-A (NTP Server GUI)

```
Services Tab → NTP → Enable Authentication
Key: 1 | Password: NTPpa55
```

Step 2 – Configure R1, R2, R3 as NTP Clients

R1

```
R1(config)# ntp server 192.168.1.5
```

R2

```
R2(config)# ntp server 192.168.1.5
```

R3

```
R3(config)# ntp server 192.168.1.5
```

> Verify with: `show ntp status`

Step 3 – Update Hardware Clock from NTP

R1 / R2 / R3

```
R1(config)# ntp update-calendar
R2(config)# ntp update-calendar
R3(config)# ntp update-calendar
```

Step 4 – Configure NTP Authentication on Routers (Key 1 · Password: NTPpa55)

R1

```
R1(config)# ntp authenticate
R1(config)# ntp trusted-key 1
R1(config)# ntp authentication-key 1 md5 NTPpa55
```

R2

```
R2(config)# ntp authenticate
```

```
R2(config)# ntp trusted-key 1
R2(config)# ntp authentication-key 1 md5 NTPpa55
```

R3

```
R3(config)# ntp authenticate
R3(config)# ntp trusted-key 1
R3(config)# ntp authentication-key 1 md5 NTPpa55
```

Step 5 – Configure Timestamp Service for Logging

R1 / R2 / R3

```
R1(config)# service timestamps log datetime msec
R2(config)# service timestamps log datetime msec
R3(config)# service timestamps log datetime msec
```

Part 3 – Configure Syslog Logging (Syslog Server: 192.168.1.6)

R1 / R2 / R3

```
R1(config)# logging host 192.168.1.6
R2(config)# logging host 192.168.1.6
R3(config)# logging host 192.168.1.6
```

> Verify with: `show logging`

Part 4 – Configure SSH on R3

Steps 1–7 – Full SSH Configuration on R3

R3

```
R3(config)# ip domain-name ccnasecurity.com
R3(config)# username SSHadmin privilege 15 secret ciscosshpa55
R3(config)# line vty 0 4
R3(config-line)# login local
R3(config-line)# transport input ssh
R3(config)# crypto key zeroize rsa
R3(config)# crypto key generate rsa
! When prompted → enter: 1024
R3(config)# ip ssh time-out 90
R3(config)# ip ssh authentication-retries 2
R3(config)# ip ssh version 2
```

Step 8–10 – Test SSH Connections

PC-C → SSH to R3 (192.168.3.1)

```
PC> ssh -l SSHadmin 192.168.3.1
Password: ciscosshpa55
```

R2 → SSH to R3 via S0/0/1 (10.2.2.1)

```
R2# ssh -v 2 -l SSHadmin 10.2.2.1
Password: ciscosshpa55
```

> Telnet to R3 should FAIL (only SSH is permitted on VTY lines).

Practical 2 – Configure AAA Authentication on Cisco Routers

R1 (Local AAA) · R2 (TACACS+ Server: 192.168.2.2) · R3 (RADIUS Server: 192.168.3.2)

Objectives

- Configure Local AAA on R1 for Console and VTY access
- Configure TACACS+ server-based AAA on R2
- Configure RADIUS server-based AAA on R3

Part 1 – Local AAA for Console Access on R1

R1

```
R1(config)# username Admin1 secret admin1pa55
R1(config)# aaa new-model
R1(config)# aaa authentication login default local
R1(config)# line console 0
R1(config-line)# login authentication default
```

Part 2 – Local AAA for VTY Lines (SSH) on R1

R1

```
R1(config)# ip domain-name ccnasecurity.com
R1(config)# crypto key generate rsa
! When prompted → enter: 1024
R1(config)# aaa authentication login SSH-LOGIN local
R1(config)# line vty 0 4
R1(config-line)# login authentication SSH-LOGIN
R1(config-line)# transport input ssh
```

Verify SSH from PC-A to R1

PC-A

```
PC> ssh -l Admin1 192.168.1.1
Password: admin1pa55
```

Part 3 – TACACS+ Server-Based AAA on R2

R2

```
R2(config)# username Admin2 secret admin2pa55
R2(config)# tacacs-server host 192.168.2.2
R2(config)# tacacs-server key tacacspa55
R2(config)# aaa new-model
R2(config)# aaa authentication login default group tacacs+ local
R2(config)# line console 0
R2(config-line)# login authentication default
```

Login with: Username: **Admin2** | Password: **admin2pa55**

Part 4 – RADIUS Server-Based AAA on R3

R3

```
R3(config)# username Admin3 secret admin3pa55
R3(config)# radius-server host 192.168.3.2
R3(config)# radius-server key radiuspa55
R3(config)# aaa new-model
R3(config)# aaa authentication login default group radius local
R3(config)# line console 0
R3(config-line)# login authentication default
```

Login with: Username: **Admin3** | Password: **admin3pa55**

Practical 3 (Scenario 1) – Configuring Extended ACLs

Router: R1 · PC1 (172.22.34.66) · PC2 (172.22.34.98) · Server (172.22.34.62)

Objectives

- PC1: FTP + ICMP access to Server only
- PC2: HTTP + ICMP access to Server only
- PCs cannot ping each other

Part 1 – Extended Numbered ACL 100 (PC1 LAN → Server: FTP + ICMP)

R1

```
R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
R1(config)# access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
! All other traffic denied by implicit deny
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip access-group 100 in
```

Wildcard for /27: 255.255.255.224 → 0.0.0.31

Part 2 – Extended Named ACL: HTTP_ONLY (PC2 LAN → Server: HTTP + ICMP)

R1

```
R1(config)# ip access-list extended HTTP_ONLY
R1(config-ext-nacl)# permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
R1(config-ext-nacl)# permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
R1(config-ext-nacl)# exit
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip access-group HTTP_ONLY in
```

Wildcard for /28: 255.255.255.240 → 0.0.0.15

Practical 3 (Scenario 2) – Configuring Extended ACLs

Router: RTA · PCA (10.101.117.51) · PCB (10.101.117.35) · SWC VLAN1 (10.101.117.2)

Policy

- SSH from 10.101.117.32/28 network → allowed to 10.101.117.0/27 network
- ICMP from any source → allowed to any destination
- All other traffic to 10.101.117.0/27 → blocked

Part 1 – Extended Numbered ACL 199 (Applied Outbound on G0/2)

RTA

```
RTA(config)# access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22
RTA(config)# access-list 199 permit icmp any any
! All other traffic denied implicitly
RTA(config)# interface gigabitEthernet 0/2
RTA(config-if)# ip access-group 199 out
```

SSH test from PCB: `PC> ssh -l Admin 10.101.117.2` (Password: Adminpa55) — should SUCCEED.
SSH from PCA directly to SWC — should FAIL. SSH PCA → SWB → SWC — will SUCCEED (bypasses ACL).

Practical 4 – Configure IP ACLs to Mitigate Attacks

R1, R2, R3 · PC-A (Server: 192.168.1.3) · PC-C (Management: 192.168.3.3)

Objectives

- Restrict SSH/Telnet access to routers to PC-C only (ACL 10)
- Permit DNS, SMTP, FTP to PC-A; deny HTTPS; allow PC-C SSH to R1 (ACL 120 on R1 S0/0/0)
- Allow only internal LAN traffic outbound on R3 G0/1 (ACL 110)
- Block RFC1918 / 127.x / Multicast on R3 S0/0/1 (ACL 100)

Part 2 – ACL 10: Restrict VTY Access to PC-C Only

R1

```
R1(config)# access-list 10 permit host 192.168.3.3
R1(config)# line vty 0 4
R1(config-line)# access-class 10 in
```

R2

```
R2(config)# access-list 10 permit host 192.168.3.3
R2(config)# line vty 0 4
R2(config-line)# access-class 10 in
```

R3

```
R3(config)# access-list 10 permit host 192.168.3.3
R3(config)# line vty 0 4
R3(config-line)# access-class 10 in
```

Part 3 – ACL 120 on R1 S0/0/0 (Inbound): Permit DNS/SMTP/FTP to PC-A, Deny HTTPS, Allow SSH from PC-C

R1

```
R1(config)# access-list 120 permit udp any host 192.168.1.3 eq domain
R1(config)# access-list 120 permit tcp any host 192.168.1.3 eq smtp
R1(config)# access-list 120 permit tcp any host 192.168.1.3 eq ftp
R1(config)# access-list 120 deny tcp any host 192.168.1.3 eq 443
R1(config)# access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22
R1(config)# interface s0/0/0
R1(config-if)# ip access-group 120 in
```

Part 4 – Modify ACL 120: Permit ICMP Echo-Reply and Unreachable, Deny All Other ICMP

R1

```
R1(config)# access-list 120 permit icmp any any echo-reply
R1(config)# access-list 120 permit icmp any any unreachable
R1(config)# access-list 120 deny icmp any any
R1(config)# access-list 120 permit ip any any
```

Part 5 – ACL 110 on R3 G0/1 (Inbound): Allow Only Internal Traffic Outbound

R3

```
R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 any
R3(config)# interface g0/1
R3(config-if)# ip access-group 110 in
```

Part 6 – ACL 100 on R3 S0/0/1 (Inbound): Block Bogon/RFC1918/Multicast Addresses

R3

```
R3(config)# access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
R3(config)# access-list 100 deny ip 10.0.0.0 0.255.255.255 any
R3(config)# access-list 100 deny ip 172.16.0.0 0.15.255.255 any
R3(config)# access-list 100 deny ip 192.168.0.0 0.0.255.255 any
R3(config)# access-list 100 deny ip 127.0.0.0 0.255.255.255 any
R3(config)# access-list 100 deny ip 224.0.0.0 15.255.255.255 any
R3(config)# access-list 100 permit ip any any
R3(config)# interface s0/0/1
R3(config-if)# ip access-group 100 in
```

Practical 5 – Configuring IPv6 ACLs

R1 (BLOCK_HTTP on G0/1 inbound) · R3 (BLOCK_ICMP on G0/0 outbound) · Server3: 2001:DB8:1:30::30/64

Objectives

- Block HTTP/HTTPS from 2001:DB8:1:11::/64 (PC2 network) to Server3
- Block all ICMP pings to Server3

Part 1 – ACL BLOCK_HTTP on R1 (Block HTTP/HTTPS from PC2's network)

R1

```
R1(config)# ipv6 access-list BLOCK_HTTP
R1(config-ipv6-acl)# deny tcp any host 2001:DB8:1:30::30 eq www
R1(config-ipv6-acl)# deny tcp any host 2001:DB8:1:30::30 eq 443
R1(config-ipv6-acl)# permit ipv6 any any
R1(config)# interface GigabitEthernet0/1
R1(config-if)# ipv6 traffic-filter BLOCK_HTTP in
```

PC1 (different network) can still browse Server3. PC2 browsing is blocked. Pings still work from PC2.

Part 2 – ACL BLOCK_ICMP on R3 (Block All ICMP to Server3)

R3

```
R3(config)# ipv6 access-list BLOCK_ICMP
R3(config-ipv6-acl)# deny icmp any any
R3(config-ipv6-acl)# permit ipv6 any any
R3(config)# interface GigabitEthernet0/0
R3(config-if)# ipv6 traffic-filter BLOCK_ICMP out
```

After this, pings from both PC1 and PC2 to Server3 will FAIL. HTTP from PC1 should still WORK.

Practical 6 – Configuring a Zone-Based Policy Firewall (ZPF)

R3 (ZPF configured) · IN-ZONE: G0/1 (192.168.3.x) · OUT-ZONE: S0/0/1

Objectives

- Internal hosts (PC-C) can initiate outbound traffic
- External hosts (PC-A) cannot initiate traffic to internal network

Part 2 – Create Zones on R3

R3

```
R3(config)# license boot module c1900 technology-package securityk9
! Accept EULA, save and reload
R3(config)# zone security IN-ZONE
R3(config-sec-zone)# exit
R3(config)# zone security OUT-ZONE
R3(config-sec-zone)# exit
```

Part 3 – Create Class-Map to Identify Internal Traffic

R3

```
R3(config)# access-list 101 permit ip 192.168.3.0 0.0.0.255 any
R3(config)# class-map type inspect match-all IN-NET-CLASS-MAP
R3(config-cmap)# match access-group 101
R3(config-cmap)# exit
```

Part 4 – Specify Firewall Policy

R3

```
R3(config)# policy-map type inspect IN-2-OUT-PMAP
R3(config-pmap)# class type inspect IN-NET-CLASS-MAP
R3(config-pmap-c)# inspect
R3(config-pmap-c)# exit
R3(config-pmap)# exit
```

Part 5 – Apply Firewall Policies

R3

```
R3(config)# zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE
R3(config-sec-zone-pair)# service-policy type inspect IN-2-OUT-PMAP
R3(config-sec-zone-pair)# exit
R3(config)# interface g0/1
R3(config-if)# zone-member security IN-ZONE
R3(config-if)# exit
R3(config)# interface s0/0/1
R3(config-if)# zone-member security OUT-ZONE
R3(config-if)# exit
```

Verification

PC-C → ping PC-A (192.168.1.3)

```
! Should SUCCEED (outbound from IN-ZONE)
```

PC-A → ping PC-C (192.168.3.3)

```
! Should FAIL (inbound from OUT-ZONE not permitted)
```

R3

```
R3# show policy-map type inspect zone-pair sessions
```

Practical 7 – Configure IOS Intrusion Prevention System (IPS) Using CLI

R1 (IPS enabled on G0/1 outbound) · Syslog Server: 192.168.1.50

Objectives

- Enable IPS on R1 to scan traffic entering the 192.168.1.0 network
- Drop ICMP echo requests (signature 2004)
- Log IPS events to Syslog server

Part 1 – Enable IOS IPS on R1

R1

```
! Step 1: Enable Security Package
R1(config)# license boot module c1900 technology-package securityk9
! Accept EULA, save and reload

! Step 3: Create IPS directory in flash
R1# mkdir ipsdir
! Press Enter to confirm

! Step 4-6: Configure IPS location, rule, and logging
R1(config)# ip ips config location flash:ipsdir
R1(config)# ip ips name iosips
R1(config)# ip ips notify log
R1# clock set 10:20:00 10 january 2014
R1(config)# service timestamps log datetime msec
R1(config)# logging host 192.168.1.50

! Step 7: Configure signature categories
R1(config)# ip ips signature-category
R1(config-ips-category)# category all
R1(config-ips-category-action)# retired true
R1(config-ips-category-action)# exit
R1(config-ips-category)# category ios_ips basic
R1(config-ips-category-action)# retired false
R1(config-ips-category-action)# exit
R1(config-ips-category)# exit
! Press Enter to confirm changes

! Step 8: Apply IPS rule outbound on G0/1
R1(config)# interface g0/1
R1(config-if)# ip ips iosips out
```

Part 2 – Modify Signature 2004 (ICMP Echo Request → Alert + Drop)

R1

```
R1(config)# ip ips signature-definition
R1(config-sigdef)# signature 2004 0
R1(config-sigdef-sig)# status
R1(config-sigdef-sig-status)# retired false
R1(config-sigdef-sig-status)# enabled true
R1(config-sigdef-sig-status)# exit
R1(config-sigdef-sig)# engine
R1(config-sigdef-sig-engine)# event-action produce-alert
R1(config-sigdef-sig-engine)# event-action deny-packet-inline
```

```
R1(config-sigdef-sig-engine)# exit
R1(config-sigdef-sig)# exit
R1(config-sigdef)# exit
! Press Enter to confirm changes
```

Verification

PC-C → ping PC-A

```
! Should FAIL — echo request is dropped by IPS
```

PC-A → ping PC-C

```
! Should SUCCEED — IPS only blocks echo request (2004), not echo reply
```

R1

```
R1# show ip ips all
```

Practical 8 – Packet Tracer: Layer 2 Security

Switches: Central (3560), SW-1, SW-2, SW-A, SW-B · PCs: C1–C4, D1–D4

Objectives

- Set Central switch as root bridge, SW-1 as secondary
- Enable PortFast and BPDU Guard on access ports (SW-A, SW-B)
- Enable Root Guard on SW-1 and SW-2 uplink ports
- Configure Port Security on all host-facing ports
- Disable unused ports

Part 1 – Configure Root Bridge

Central

```
Central(config)# spanning-tree vlan 1 root primary
```

SW-1

```
SW-1(config)# spanning-tree vlan 1 root secondary
```

Verify with: `Central# show spanning-tree` — should show "This bridge is the root"

Part 2 – Protect Against STP Attacks

PortFast + BPDU Guard on Access Ports (SW-A, SW-B)

SW-A

```
SW-A(config)# interface range f0/1 - 4
SW-A(config-if-range)# spanning-tree portfast
SW-A(config-if-range)# spanning-tree bpduguard enable
```

SW-B

```
SW-B(config)# interface range f0/1 - 4
SW-B(config-if-range)# spanning-tree portfast
SW-B(config-if-range)# spanning-tree bpduguard enable
```

Root Guard on SW-1 and SW-2 (Ports F0/23–24)

SW-1

```
SW-1(config)# interface range f0/23 - 24
SW-1(config-if-range)# spanning-tree guard root
```

SW-2

```
SW-2(config)# interface range f0/23 - 24
SW-2(config-if-range)# spanning-tree guard root
```

Part 3 – Port Security and Disable Unused Ports

SW-A

```
SW-A(config)# interface range f0/1 - 22
SW-A(config-if-range)# switchport mode access
SW-A(config-if-range)# switchport port-security
SW-A(config-if-range)# switchport port-security maximum 2
SW-A(config-if-range)# switchport port-security violation shutdown
SW-A(config-if-range)# switchport port-security mac-address sticky
SW-A(config)# interface range f0/5 - 22
SW-A(config-if-range)# shutdown
```

SW-B

```
SW-B(config)# interface range f0/1 - 22
SW-B(config-if-range)# switchport mode access
SW-B(config-if-range)# switchport port-security
SW-B(config-if-range)# switchport port-security maximum 2
SW-B(config-if-range)# switchport port-security violation shutdown
SW-B(config-if-range)# switchport port-security mac-address sticky
SW-B(config)# interface range f0/5 - 22
SW-B(config-if-range)# shutdown
```

Verify with: `SW-A# show port-security interface f0/1`

Practical 9 – Layer 2 VLAN Security

Switches: Central, SW-1, SW-2, SW-A, SW-B · Router: R1 · Management PC: 192.168.20.50

Objectives

- Add a secure trunk link between SW-1 and SW-2 (Native VLAN 15, no DTP)
- Create VLAN 20 (Management) on all switches
- Secure management VLAN from other VLANs using ACLs on R1

Part 2 – Redundant Trunk Link SW-1 ↔ SW-2 (Port F0/23)

SW-1

```
SW-1(config)# interface f0/23
SW-1(config-if)# switchport mode trunk
SW-1(config-if)# switchport trunk native vlan 15
SW-1(config-if)# switchport nonegotiate
SW-1(config-if)# no shutdown
```

SW-2

```
SW-2(config)# interface f0/23
```

```
SW-2(config-if)# switchport mode trunk
SW-2(config-if)# switchport trunk native vlan 15
SW-2(config-if)# switchport nonegotiate
SW-2(config-if)# no shutdown
```

Part 3 – Enable Management VLAN 20 on All Switches

SW-A

```
SW-A(config)# vlan 20
SW-A(config-vlan)# exit
SW-A(config)# interface vlan 20
SW-A(config-if)# ip address 192.168.20.1 255.255.255.0
SW-A(config)# interface f0/1
SW-A(config-if)# switchport access vlan 20
SW-A(config-if)# no shutdown
```

SW-B

```
SW-B(config)# vlan 20
SW-B(config-vlan)# exit
SW-B(config)# interface vlan 20
SW-B(config-if)# ip address 192.168.20.2 255.255.255.0
```

SW-1

```
SW-1(config)# vlan 20
SW-1(config-vlan)# exit
SW-1(config)# interface vlan 20
SW-1(config-if)# ip address 192.168.20.3 255.255.255.0
```

SW-2

```
SW-2(config)# vlan 20
SW-2(config-vlan)# exit
SW-2(config)# interface vlan 20
SW-2(config-if)# ip address 192.168.20.4 255.255.255.0
```

Central

```
Central(config)# vlan 20
Central(config-vlan)# exit
Central(config)# interface vlan 20
Central(config-if)# ip address 192.168.20.5 255.255.255.0
```

Part 4 – Enable Management PC Access to Router R1 + ACL Security

R1

```
! Subinterface for VLAN 20
R1(config)# interface g0/0.3
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 192.168.20.100 255.255.255.0

! ACL 101: Block all traffic destined for VLAN 20 from other VLANs
R1(config)# access-list 101 deny ip any 192.168.20.0 0.0.0.255
R1(config)# access-list 101 permit ip any any

! ACL 102: Allow only Management PC SSH to the router
R1(config)# access-list 102 permit ip host 192.168.20.50 any

! Apply ACLs
R1(config)# interface g0/0.1
R1(config-subif)# ip access-group 101 in
R1(config)# interface g0/0.2
```

```
R1(config-subif)# ip access-group 101 in
R1(config)# line vty 0 4
R1(config-line)# access-class 102 in
```

Verify SSH from Management PC to R1

Management PC

```
PC> ssh -l SSHadmin 192.168.20.100
Password: ciscosshpa55
```

Practical 10 – Configure and Verify a Site-to-Site IPsec VPN Using CLI

VPN Tunnel: R1 (192.168.1.x) ↔ R3 (192.168.3.x) via R2 (pass-through)

VPN Parameters Summary

- IKE Phase 1: AES-256, SHA-1, Pre-share, DH Group 5, Lifetime 86400s, Key: vpnpa55
- IKE Phase 2: Transform-Set VPN-SET (esp-aes + esp-sha-hmac), Crypto Map VPN-MAP
- R1 peers to 10.2.2.2 (R3's S0/0/1) · R3 peers to 10.1.1.2 (R1's S0/0/0)

Part 1 – Configure IPsec on R1

R1

```
! Step 2: Enable Security Package
R1(config)# license boot module c1900 technology-package securityk9
! Accept EULA, save and reload

! Step 3: Identify interesting traffic (LAN to LAN)
R1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255

! Step 4: IKE Phase 1 - ISAKMP Policy
R1(config)# crypto isakmp policy 10
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 5
R1(config-isakmp)# exit
R1(config)# crypto isakmp key vpnpa55 address 10.2.2.2

! Step 5: IKE Phase 2 - Transform Set + Crypto Map
R1(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)# crypto map VPN-MAP 10 ipsec-isakmp
R1(config-crypto-map)# description VPN connection to R3
R1(config-crypto-map)# set peer 10.2.2.2
R1(config-crypto-map)# set transform-set VPN-SET
R1(config-crypto-map)# match address 110
R1(config-crypto-map)# exit

! Step 6: Apply crypto map to outgoing interface
R1(config)# interface s0/0/0
R1(config-if)# crypto map VPN-MAP
```

Part 2 – Configure IPsec on R3

R3

```
! Step 2: Interesting traffic (mirror of R1)
R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255

! Step 3: ISAKMP Policy
R3(config)# crypto isakmp policy 10
```

```
R3(config-isakmp)# encryption aes 256
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)# group 5
R3(config-isakmp)# exit
R3(config)# crypto isakmp key vpnpa55 address 10.1.1.2

! Step 4: Transform Set + Crypto Map
R3(config)# crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R3(config)# crypto map VPN-MAP 10 ipsec-isakmp
R3(config-crypto-map)# description VPN connection to R1
R3(config-crypto-map)# set peer 10.1.1.2
R3(config-crypto-map)# set transform-set VPN-SET
R3(config-crypto-map)# match address 110
R3(config-crypto-map)# exit

! Step 5: Apply to outgoing interface
R3(config)# interface s0/0/1
R3(config-if)# crypto map VPN-MAP
```

Part 3 – Verify IPsec VPN

R1

```
R1# show crypto ipsec sa
! Before traffic: encaps/decaps = 0
! After ping PC-A → PC-C: encaps/decaps > 0 (tunnel working)
! After ping PC-A → PC-B (uninteresting): packet count unchanged
```

PC-A → ping PC-C (Interesting Traffic)

```
PC> ping 192.168.3.3
! This triggers the VPN tunnel
```

Practical 11 – Configuring ASA Basic Settings and Firewall Using CLI

ASA 5505 · Inside VLAN1: 192.168.1.x · Outside VLAN2: 209.165.200.x · DMZ VLAN3: 192.168.2.x

Objectives

- Configure ASA hostname, domain, passwords, interfaces with security levels
- Configure static default route and PAT (NAT)
- Configure MPF (ICMP inspection), DHCP, AAA, SSH
- Configure DMZ with static NAT and ACLs

Part 2 – Basic ASA Settings and Interface Security

ASA

```
ASA(config)# hostname CCNAS-ASA
CCNAS-ASA(config)# domain-name ccnasecurity.com
CCNAS-ASA(config)# enable password ciscoenpa55

! VLAN 1 - Inside (Security Level 100)
CCNAS-ASA(config)# interface vlan 1
CCNAS-ASA(config-if)# nameif inside
CCNAS-ASA(config-if)# ip address 192.168.1.1 255.255.255.0
CCNAS-ASA(config-if)# security-level 100

! VLAN 2 - Outside (Security Level 0)
CCNAS-ASA(config-if)# interface vlan 2
CCNAS-ASA(config-if)# nameif outside
```

```
CCNAS-ASA(config-if)# ip address 209.165.200.226 255.255.255.248
CCNAS-ASA(config-if)# security-level 0
```

Part 3 – Routing, PAT, and ICMP Inspection

ASA

```
! Static default route via R1 G0/0
CCNAS-ASA(config)# route outside 0.0.0.0 0.0.0.0 209.165.200.225

! PAT using network object
CCNAS-ASA(config)# object network inside-net
CCNAS-ASA(config-network-object)# subnet 192.168.1.0 255.255.255.0
CCNAS-ASA(config-network-object)# nat (inside,outside) dynamic interface
CCNAS-ASA(config-network-object)# end

! MPF - Inspect ICMP
CCNAS-ASA(config)# class-map inspection_default
CCNAS-ASA(config-cmap)# match default-inspection-traffic
CCNAS-ASA(config-cmap)# exit
CCNAS-ASA(config)# policy-map global_policy
CCNAS-ASA(config-pmap)# class inspection_default
CCNAS-ASA(config-pmap-c)# inspect icmp
CCNAS-ASA(config-pmap-c)# exit
CCNAS-ASA(config)# service-policy global_policy global
```

Part 4 – DHCP, AAA, and SSH

ASA

```
! DHCP Server for inside network
CCNAS-ASA(config)# dhcpd address 192.168.1.5-192.168.1.36 inside
CCNAS-ASA(config)# dhcpd dns 209.165.201.2 interface inside
CCNAS-ASA(config)# dhcpd enable inside

! Local user + AAA authentication
CCNAS-ASA(config)# username admin password adminpa55
CCNAS-ASA(config)# aaa authentication ssh console LOCAL

! RSA key + SSH access rules
CCNAS-ASA(config)# crypto key generate rsa modulus 1024
! If prompted to replace existing keys → enter: no
CCNAS-ASA(config)# ssh 192.168.1.0 255.255.255.0 inside
CCNAS-ASA(config)# ssh 172.16.3.3 255.255.255.255 outside
CCNAS-ASA(config)# ssh timeout 10
```

Test SSH Connections

PC-C → SSH to ASA (Outside)

```
PC> ssh -l admin 209.165.200.226
Password: adminpa55
```

PC-B → SSH to ASA (Inside)

```
PC> ssh -l admin 192.168.1.1
Password: adminpa55
```

Part 5 – Configure DMZ (VLAN 3), Static NAT, and ACLs

ASA

```
! VLAN 3 - DMZ (Security Level 70)
CCNAS-ASA(config)# interface vlan 3
CCNAS-ASA(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
CCNAS-ASA(config-if)# no forward interface vlan 1
CCNAS-ASA(config-if)# nameif dmz
CCNAS-ASA(config-if)# security-level 70

! Assign physical port E0/2 to VLAN 3
CCNAS-ASA(config-if)# interface Ethernet0/2
CCNAS-ASA(config-if)# switchport access vlan 3

! Static NAT for DMZ server (192.168.2.3 → 209.165.200.227)
CCNAS-ASA(config)# object network dmz-server
CCNAS-ASA(config-network-object)# host 192.168.2.3
CCNAS-ASA(config-network-object)# nat (dmz,outside) static 209.165.200.227
CCNAS-ASA(config-network-object)# exit

! ACL: Allow ICMP + HTTP from outside to DMZ server
CCNAS-ASA(config)# access-list OUTSIDE-DMZ permit icmp any host 192.168.2.3
CCNAS-ASA(config)# access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3 eq 80
CCNAS-ASA(config)# access-group OUTSIDE-DMZ in interface outside
```

> PC-B should be changed from static IP to DHCP addressing after DHCP is configured.